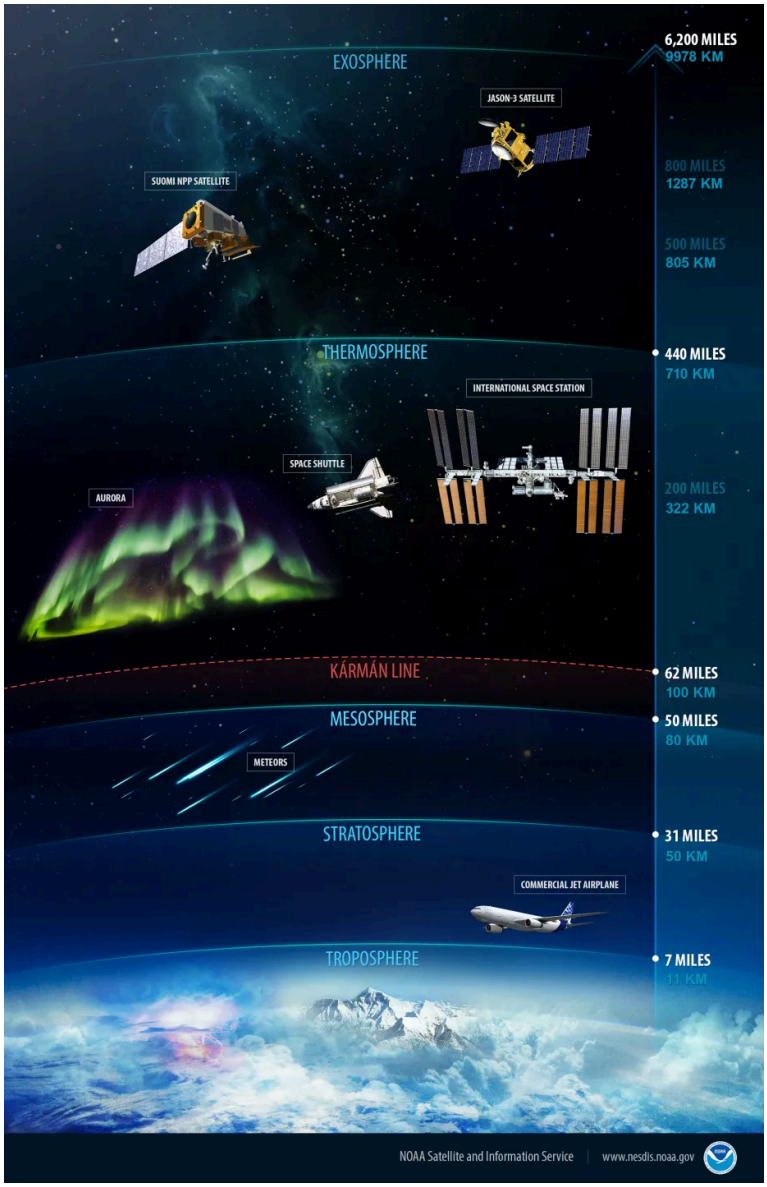
MGT4600 – MANAGEMENT OF TECHNOLOGY AND INNOVATION

MIDTERM PROJECT REPORT – GROUP 2

Samuel Darteh[1], Brandon Michalak[1], Ganesh Vannam[1], Alex Nordstrom[1] | Michigan Technological University[1]

HYPERSONIC JETS: FLYING AT HIGH ALTITUDES



1. Project Objective

The project aims to analyze the prospects of high-altitude flight technology and its impact on developments within the aviation industry and related market movements.

Problem Ideation

Transporting people and equipment by air has drawbacks, primarily due to long travel times. While factors like flight route, wind patterns, and aircraft type largely affect travel duration, situations such as emergency response, healthcare, and high-profile movements are critically impacted, inevitably. A non-stop flight for example, from New York to London averages 7-8 hours over 3,470 miles (5,585 km) translating to an average speed of 434-496 mph (698-798 km/h).[1] Notwithstanding, there has been a long standing dream of *hypersonic* transport—the dream of flying from London to Tokyo in 90 minutes.[2]

This research investigates the potential impact of hypersonic transport on the aviation industry including the technology that could support the innovation and its safety implications.

2. Characteristics of High Altitude Flights

Evolution of Flight Technology

In 1903, the Wright brothers laid the foundation for what would be considered as an architectural innovation, disrupting both technological competencies and market linkages of the transport space—moving people and equipment by air—over the options offered by the automotive and marine industries.

The 20th century witnessed critical technologies and innovations in aircraft engineering such as lightweight building materials, optimized engines, improved aerodynamics, efficient propulsion systems, and safer flight operations. After the Wright brothers breakthrough in the aerospace industry, research and development teams have aimed at always flying faster, and flying higher.

[1] Skyscanner Url: https://www.skyscanner.com/routes/nyca/lond/new-york-to-london.html. Also, FlyAware Url: https://www.flightaware.com/live/flight/BAW172/history/20241021/0105Z/KJFK/EGLL

[2] Williamson, Mark, "Hypersonic Transport … 30 Years and Holding?", Aerospace America, Vol. 50, No. 5, May 2012, pp 40-45.

Flying at High Altitude

The flyzone for all aircraft types except for space shuttles occur in the stratosphere measuring approximately 30,000-164,000 feet (7-31 miles, 11-50 km).[3] Commercial aircrafts fly between 30,000-40,000 feet. High-altitude flights like supersonic and hypersonic flights typically fly above 40,000 and 60,000 feet, over the troposphere in the stratosphere—for "thinner air."[4]

Considering speed, commercial aircrafts fly below *Mach 1*—the speed of sound at sea level, about 761 mph. Supersonic flights have speeds between *Mach 0.99-1.01* and *hypersonic* flights at minimum of *Mach 5.01*—five times the speed of sound.

> Hypersonic flight is that high-speed part of the flight spectrum where certain physical phenomena become more important that are not so important at lower speeds. One of those physical phenomena is intense aerodynamic heating of the surface due to skin friction and shock wave heating. Protection against aerodynamic heating is the dominant feature that drives the design of hypersonic vehicles.[5]

The reduced air resistance at such high altitudes aids faster speed with lesser fuel consumption, and has fewer weather disruptions for better control and smoother flights. Hybrid propulsion systems are used for these hypersonic flights for efficient fuel management. Electric motors are used for takeoff and landing while traditional fuel engines are used when space vehicles hit the target altitude to maximize performance and fuel efficiency—which mode is friendlier to the ecosystem. Examples of existing high-altitude aircrafts or vehicles include the Concorde, military jets, and stratospheric balloons.

Features of High Altitude Flights

For basic flights similar to the 1903 aircraft model, technologies not limited to the following form part of the product architecture. *Aerodynamics optimization*—for lifting, air stability and fuel efficiency; *propulsion systems*—to generate forward thrust; *lightweight frame*—using

---

[3] UCAR Center of Scientific Education: Layers of Earth's Atmosphere Url: https://scied.ucar.edu/learning-zone/atmosphere/layers-earths-atmosphere. Layers of the atmosphere have been described pictorially in the Appendix.
[4] See: Appendix for detailed description of layers of the atmosphere.
[5] Hypersonic Flights, National Air and Space Museum, Url: https://airandspace.si.edu/stories/editorial/hypersonic-flight

minimal weight materials; and *control surfaces*—ailerons, rudders, elevators, flaps, air brakes and spoilers for lifting, landing and balancing.[6]

Modern aircrafts viz-a-viz the 1903 aircraft over the past century expectedly showcase a wide gap of technological changes and groundbreaking features—such as jet engines (ramjet and Scramjet engines), fly-by-wire systems, advanced avionics, efficient-sustainable fuels, autopilot and Artificial Intelligence (AI).[7]

Standards and Dominant Designs of Hypersonic Technology

The emergent hypersonic technology builds on earlier developments, starting with the V-2 and WAC Corporal rockets in 1949, which reached altitudes of 100-244 miles and speeds of 3,500-5,150 mph. These early innovations laid the foundation for high-speed space travel. Over time, hypersonic vehicles like NASA's aircraft (X-43A), the Boom supersonic jet, and the X-15 research vehicle further advanced ultra-high-speed capabilities.

The most relevant applications for the development of modern hypersonic technology include these high-altitude flights that exceed *Mach 5* speeds which validate propulsion systems and aerodynamic designs to withstand the intense heat in the stratosphere.

Organizations now exploring the commercialization of hypersonic travel include Venus Aerospace, Velontra, Sierra Space, and Hermeus. Notably, China Space Transportation is working on a "rocket with wings" for space tourism and intercity travel. At the 2022 UP Summit, Venus Aerospace introduced the 2,000-lb VDR2 Thrust Engine, marking a pivotal step toward high-speed commercial flights.[8]

[6] Joyplanes YouTube Channel, Url:https://www.youtube.com/watch?app=desktop&v=XxrdhJanotw&t=80
[7] Jet engines: high speed propulsion and aircraft efficiency, Fly-by-wire systems: computer based flight controls, Advanced avionics: manages communication, navigation and weather, Efficient-sustainable fuels: minimizes carbon emissions, Autopilot and AI: provides better flight operations and safety.
[8] 2022 UP Summit, Url: https://www.instagram.com/stories/highlights/17947070029975807/?hl=en; Also, 2024 UP Summit, Url: https://up.partners/summit/

3. Market Implications and Analytics

Customer Expectations for High-Altitude Flights

People are always looking for ways to innovate and improve the overall performance of technology. Traditional commercial airlines have been around for quite some time and there have been advancements within the industry, but nothing that has really made it significantly faster to get from one place to another. This is the niche for hypersonic jets. Giving people the ability to cut down their travel time can be a huge advantage for any company in the aviation space.[9]

With the high cost for tickets on a hypersonic jet, customers would expect the travel times to be worth the *price* and as well ensuring the aircraft is *safe* and *comfortable*. Although the time to get to one place is much less, people will still want to feel safe traveling at high altitudes and speeds. They would also expect a *luxury* experience for the price that they would be paying. Some expectations could be *reducing time* at the airport, *comfort* within the aircraft, and high levels of *service*.

Market History and Movements

In recent decades, advancements in commercial aircraft have focused on *speed* and *efficiency*, notably with the introduction of modern engines in the 1950s and the supersonic Concorde in the 1970s. Despite its initial success, the Concorde faced commercial failure due to high costs and environmental issues. Currently, the aviation industry emphasizes *sustainability*, driving the development of fuel-efficient engines, alternative fuels, and hypersonic jets.

Key performance (market) dimensions for hypersonic jets include *speed, cost, safety,* and *environmental impact*. Speed is crucial for reducing travel times, while cost affects market acceptance. Safety is essential for regulatory approval, and a focus on environmental sustainability is increasingly important for reducing emissions and improving fuel efficiency.

In the future, hypersonic jets could transform air travel, particularly for business and emergency services. However, challenges such as *high development costs* and *regulatory hurdles* need to be

[9] See: https://www2.deloitte.com/us/en/pages/energy-and-resources/articles/rise-of-hypersonics.html. Also: https://ntrs.nasa.gov/api/citations/20210014711/downloads/Deloitte%20Consulting%20LLP%20-%20Commercial%20Hypersonic%20Transportation%20Market%20Study%20-%20April%202021.pdf

addressed. Initially, hypersonic transport may cater to niche markets, but technological advances could lead to broader adoption over time, aligning with the demand for fast, efficient, and sustainable air travel solutions.

4. Market Application: Flying at High Altitude with New Hypersonic Jets

Technology and Market Overlap: Hypersonic Jet Prototype

Hypersonic jets, unlike commercial airliners, can fly at speeds over *Mach 5* (3,800 mph), significantly faster than the typical *Mach 0.85* (560 mph) of planes like the Boeing 777. Some experimental jets, like NASA's X-43A, have reached nearly *Mach 10* (7,600 mph). This speed could reduce a New York to London flight from 7-8 hours to just 1-2 hours, making hypersonic jets ideal for long-haul flights where time is crucial.

Hypersonic jets also fly at higher altitudes—up to 100,000 feet—where *air resistance is lower*, allowing for greater speeds and reduced congestion. In contrast, commercial jets operate between 30,000 and 40,000 feet. The high altitude of hypersonic jets presents challenges in cabin pressurization and safety but enables more efficient flight at extreme speeds.

Despite their speed, hypersonic jets are *less fuel-efficient* than commercial aircraft, particularly during takeoff and acceleration. Their propulsion systems consume significantly more fuel, whereas commercial jets are optimized for fuel efficiency at cruising speeds, making them more economical for long-distance flights. One area that may be considered is hydrogen fuel, and air-breathing systems employed in these speed-focused aircrafts.[10] Hypersonic jets will need improvements in propulsion technology to become more sustainable.

Hypersonic jets are expected to have *lower passenger capacities* compared to commercial airliners, which can carry hundreds of passengers. Early hypersonic designs *focus on premium markets*, while commercial jets are optimized for mass transport and passenger comfort. Commercial aircraft provide greater space and amenities, making them more suitable for long-haul travel for large groups.

[10] One relevant project is Destinus' hydrogen-powered aircraft (Swiss start-up company), designed to operate at high altitudes to minimize drag and environmental impact. These futuristic jets aim to combine rocket and air-breathing engines, allowing speeds far beyond Mach 5. See: https://thenextweb.com/news/hypersonic-hydrogen-jet-destinus

Benchmarking: From New York to London in 60 Minutes: Hypersonic Jet Set for Test Flight in 2025[11]

The long-awaited "London-to-Tokyo 90-minute flight" dream[12] is set to become a reality through a partnership between Venus Aerospace and Velontra, with the inaugural flight planned for 2025, covering "New York to London in 60 minutes." The hypersonic jet will reach speeds of *Mach 6* (3,600 mph / 5,795 km/h), using conventional engines for takeoff and landing and switching to rocket propulsion at high altitudes, offering passengers stunning views of the Earth's curvature and outer space.

Though the hypersonic project faces operational, safety, and environmental challenges, the new technology and emerging market present compelling advantages to attract enthusiastic *early adopters*, visionary *early majority*, and even pragmatic *late majority stakeholders*. Key market features—such as speed, high-altitude stability, unique experience, future potential, and pioneer advantage—provide Venus aerospace and Velontra with unmatched competitive advantages.

With the ability to significantly reduce travel times, hypersonic jets will initially target affluent business travelers and government officials, providing premium services at high ticket prices. The reduced travel time will enhance efficiency, allowing quicker recovery from jet lag and facilitating last-minute meetings. Though currently aimed at the luxury market, hypersonic travel could eventually become more accessible as technology advances, potentially transforming the airline industry over time.

5. Analysis and Conclusion: Future Work and Prospects

Significant progress in hypersonic technology has been made through experimental and military applications. NASA's X-43A reached *Mach 9.6* in 2004, and Boeing's X-51 Waverider achieved *Mach 5.1* in 2013. Countries like Russia and China have developed operational hypersonic weapons including Russia's Avangard and China's DF-ZF showcasing military capabilities. Furthermore, companies like Virgin Galactic and Stratolaunch are exploring commercial hypersonic flight, reflecting increasing interest in reducing travel times.

---

[11] Url: https://www.ndtv.com/world-news/from-new-york-to-london-in-60-minutes-hypersonic-jet-set-for-test-flight-in-2025-6736979

[12] See: https://www.timeout.com/usa/news/can-we-really-fly-from-nyc-to-tokyo-in-one-hour-050223

The quest for faster flight, and higher flight—hypersonic travel—has not been without some failed attempts, and consequently inevitable challenges. For example:

> After the first flight attempt in June of 2001 failed when the booster rocket went out of control, the second and third attempts resulted in highly successful, record-breaking flights. *Mach 6.8* was reached in March of 2004, and Mach 9.6 was reached in the final flight in November of 2004.[13]

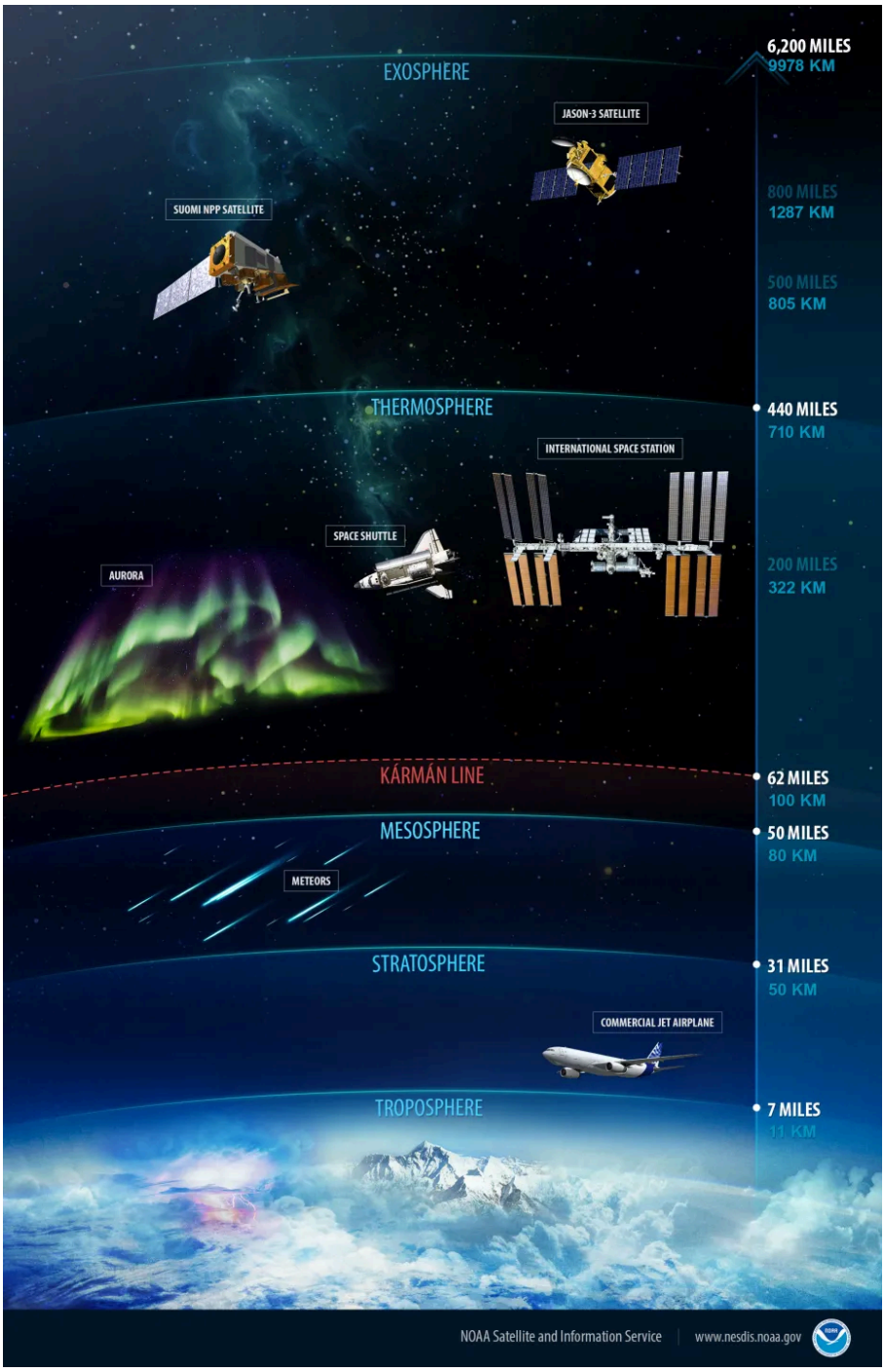
The failed missions and successful hypersonic flights to the stratosphere open opportunities for critical analysis and research into key questions that need to be addressed:

i. *What are the safety and regulatory challenges associated with hypersonic flight?*
ii. *How will hypersonic technology impact environmental sustainability?*
iii. *What reforms must be necessitated within the aviation space to make room for commercialization of hypersonic travel?*
iv. *Finally, what infrastructure changes are necessary to support hypersonic travel?*

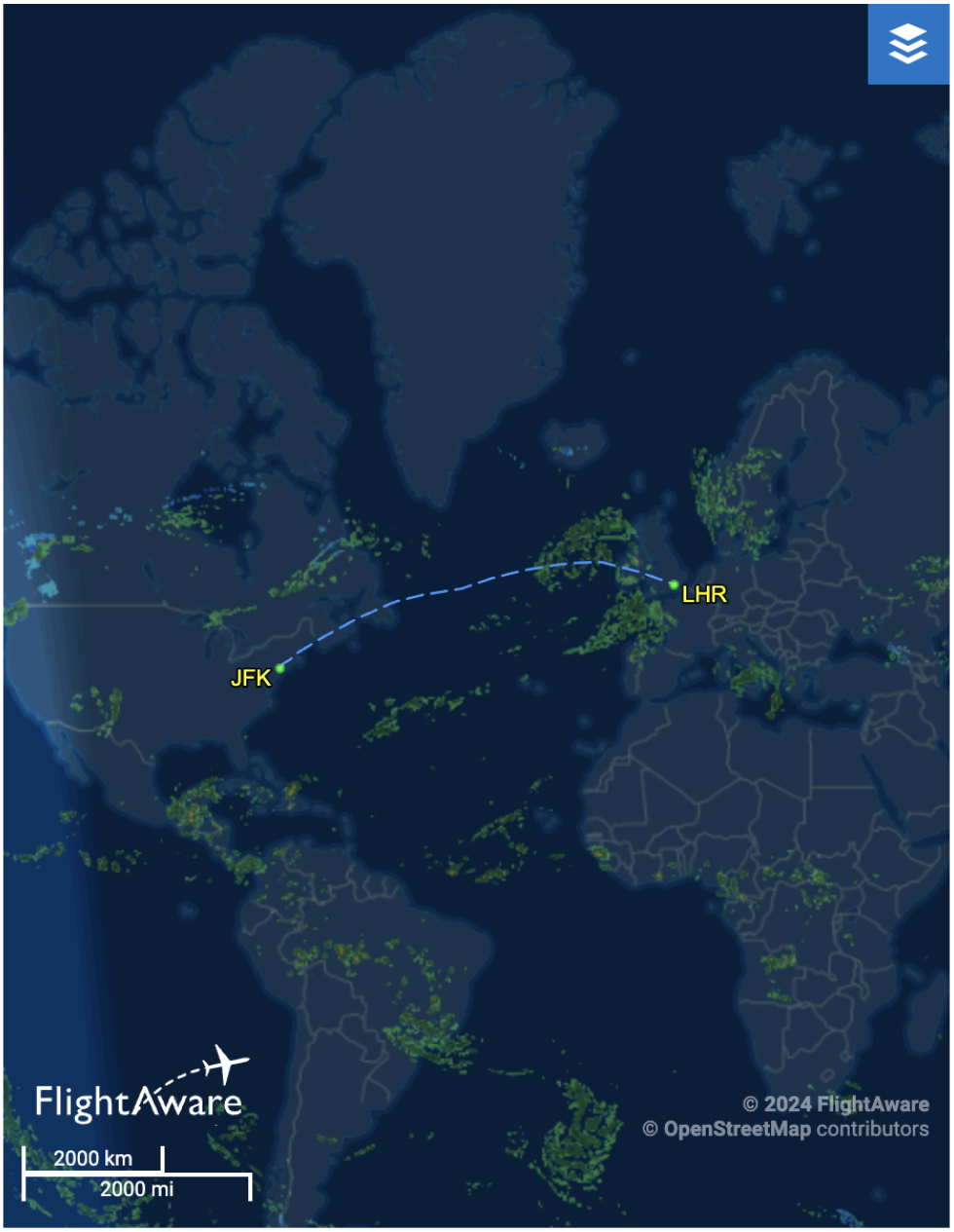
Some experts have predicted that commercial hypersonic flights could become a reality within the next decade, estimating operations as early as the 2030s.[14] While the timeline depends on overcoming technological, regulatory, safety, and infrastructural challenges, hypersonic flights promise to revolutionize air travel by drastically reducing intercontinental travel times and offering passengers breathtaking views of the sky—experiencing the troposphere, stratosphere, and mesosphere—in an unforgettable journey.

[13] NASA's X-43A subsequent flights in 2004. Url: https://www.nasa.gov/reference/x-43a/
[14] NASA Technical Reports Server: https://ntrs.nasa.gov/. See also: https://guidehouseinsights.com

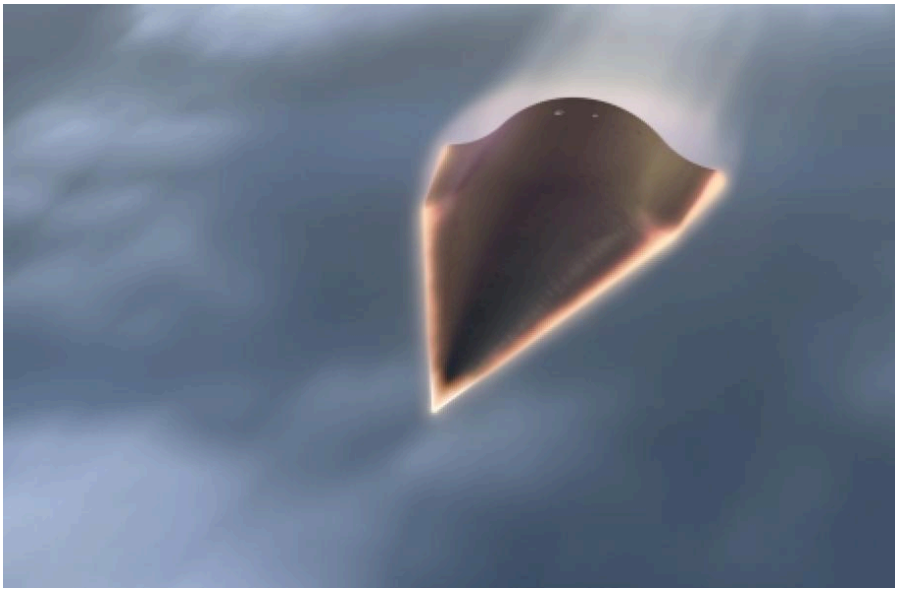
APPENDIX



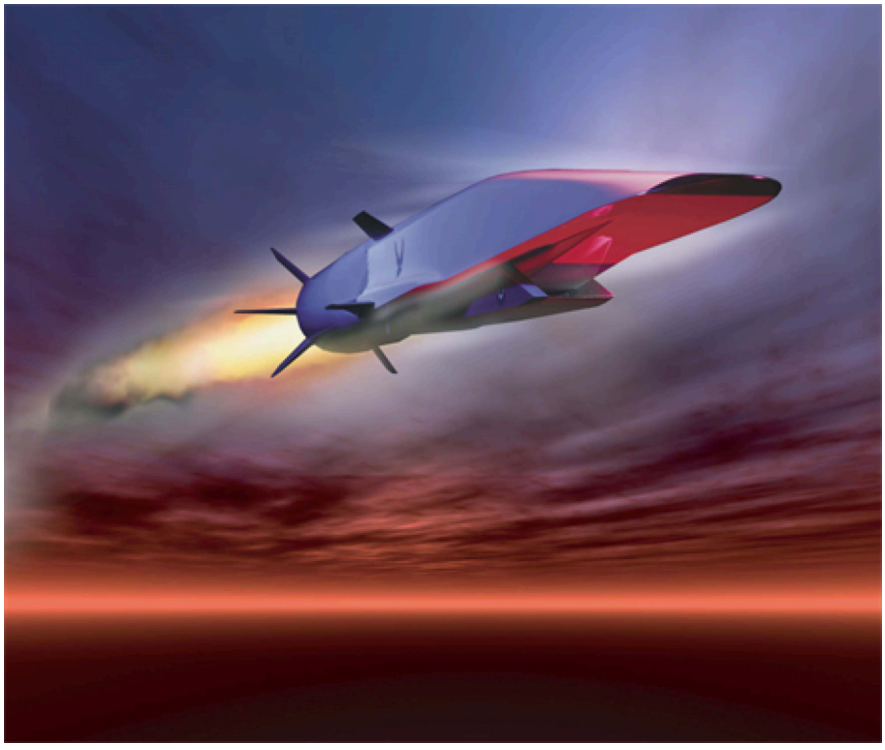
Various layers of the earth's atmosphere in vertical orientation from ground level: troposphere through the exosphere. Image from NOAA–National Oceanic and Atmospheric Administration



FlyAware Map showing the benchmarked route from New York to London for earmarked Hypersonic Jet flight test in 2025



The Falcon Hypersonic Technology Vehicle 2 (HTV-2) was announced by DARPA in 2011. Rocket surfaces peeled off after 200s at Mach 20 due to intense aerodynamic heating.



The X-51, which operated for about 200 seconds at Mach 5



The first X-43A hypersonic research aircraft was launched by the Pegasus booster rocket in June 2001



The third X-43A hypersonic research aircraft, attached to a modified Pegasus booster rocket, was taken to launch altitude by NASA's B-52B launch aircraft from the NASA Dryden Flight Research Center at Edwards, CA, on November 16, 2004.